Analysis of Air Quality Data Using Sampling and Descriptive Statistics

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INTRODUCTION

Problem
Air-quality datasets contain large numbers of observations. measurements are difficult to interpret directly. Statistical summaries help identify typical levels.

Approach
Sampling + Descriptive Statistics
- Select representative observations
- Calculate statistical measures
- Visualize distributions
- Interpret patterns

Why this study?
- Understand pollution levels
- Quantify variability
- Identify unusual observations
- Support environmental monitoring

To Calculate the statistical measures

DATASET

Date	PM2.5	PM10	NO_2	CO	O_3
Jan 1	42	78	31	0.8	25
Jan 2	35	69	28	0.7	30
Jan 3	51	91	35	0.9	21

Dataset Characteristics
- **Data source:** Air Quality Monitoring Dataset
- **Study location:** New Delhi, India
- **Study period:** January 2024 – December 2024
- **Total observations:** 10,000
- **Sample size:** 500 observations
- **Sampling method:** Simple Random Sampling
- **Variables analysed:** $PM_{2.5}$, PM_{10}, NO_2, CO, O_3

METHOD PIPELINE

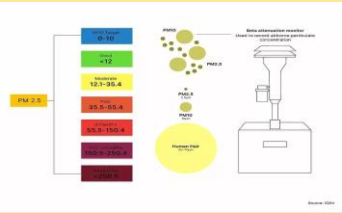
Raw Data → Cleaning → Sampling → Statistics → Visualization → Interpretation

SAMPLING OVERVIEW

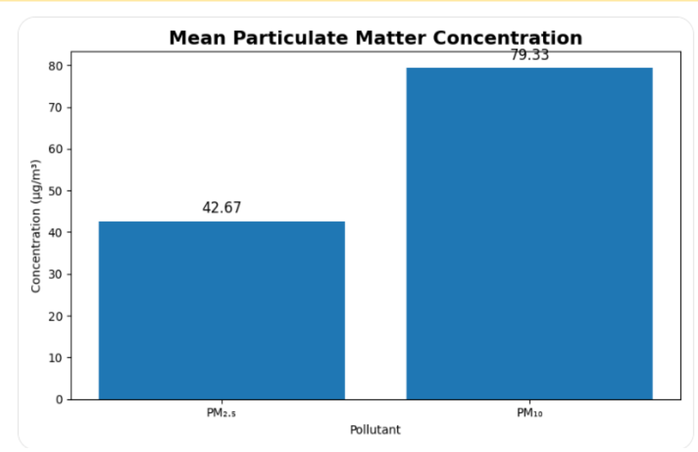
POPULATION 10,000 observations → **Sampling Method** → **SAMPLE 500 observations** → **Statistical Analysis**

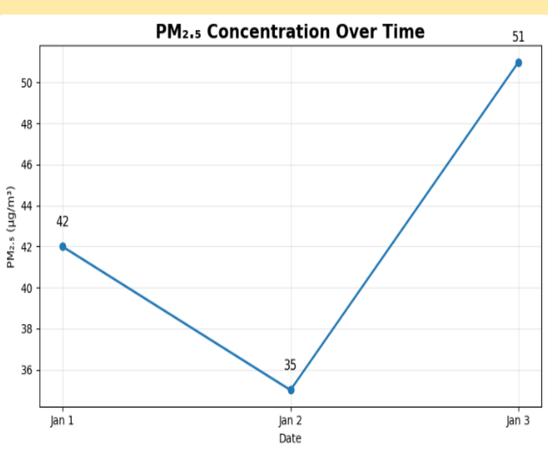
CONCLUSION

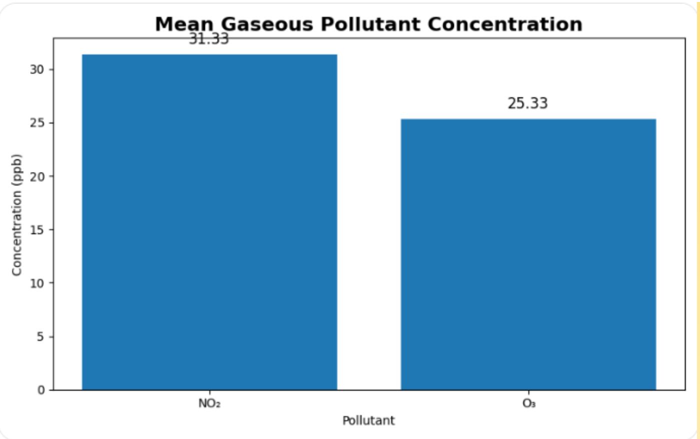
Descriptive statistics summarized the air-quality data.
PM_{10} and $PM_{2.5}$ had higher average concentrations.
Standard deviation showed differences in pollutant variability.
Sampling and visualization revealed pollution patterns.



RESULTS







FUTURE DIRECTIONS

- Increase the **sample size**.
- Study **seasonal variations**.
- Compare **different locations**.
- Include **weather factors**.
- Use **predictive models**
- **Collect Data** over a long time period
- **Incorporate** Data Visualization.